

Principle of Interiorized Probability Optimization (Possible Formalization)

The goal here is to showcase a formalized mathematical framework that justifies the scalable implementability and effectiveness of an AI architectural model that utilizes some kind of multitude-of-experts (such as mixture of experts or autonomous architecture generation) framework. In doing so, we should thus be able to create an autoregressive LLM that utilizes autonomous architecture generation.

We begin with an ontological statement:

THE PROPER DATASET FOR A MACHINE LEARNING MODEL SHOULD NOT JUST INCLUDE EXTERNAL DATA SUCH AS IMAGES OR AUDIO OR TEXT, BUT SHOULD ALSO INCLUDE THE DATA REPRESENTING THE INTERNAL ARCHITECTURAL STRUCTURE UTILIZED FOR DATA PROCESSING.

Formally, this means that if external data can be represented via a vector X_n and internal data be represented via a vector I_n , we are aiming to learn an externality-internality joint distribution $P(X_{n+1}, I_{n+1} | X_n, I_n)$. This is in contrast to regular models that are just trying to learn $P(X_{n+1} | X_n)$.

As such, when one is trying to generate the next external datapiece X_{n+1} , the idea is that representing it as sampling from a distribution conditionally dependent on I_n means we are using some expert determined by I_n to perform our computation ($[X_{n+1}, I_{n+1}] = f_{I_n}([X_n, I_n])$).

There are two big things needed to make sure this works. For one, learning $P(X_{n+1}, I_{n+1} | X_n, I_n)$ should organically allow us to learn to generate the proper external data (properly predict the next X_{n+1} in a sequence). For two, we need to be able to determine how we can even train on interiority data when it does not even exist (we have plenty of images to train on but basically no AI interiority scans to learn from reliably).

LEARNING THE EXTERNAL DATA

As a consequence of maximum likelihood estimation proofs, we know that in regular models, we can learn $P(X_{n+1} | X_n)$ by minimizing a negative log likelihood $-\log(P(X_{n+1} | X_n))$. As such, for a given sequence of interiorities I_n (assume for now that they are given), we can likewise state that we can learn $P(X_{n+1}, I_{n+1} | X_n, I_n)$ by minimizing the negative log likelihood $-\log(P(X_{n+1}, I_{n+1} | X_n, I_n))$.

However, by conditional probability rules, we know that we can do some basic statistical manipulation to get $P(X_{n+1}, I_{n+1} | X_n, I_n) = P(X_{n+1} | X_n, I_n) P(I_{n+1} | X_{n+1}, X_n, I_n)$, thus meaning that $-\log(P(X_{n+1}, I_{n+1} | X_n, I_n)) = -\log(P(X_{n+1} | X_n, I_n)) - \log(P(I_{n+1} | X_{n+1}, X_n, I_n))$. By MLE proofs, we know that $-\log(P(X_{n+1}, I_{n+1} | X_n, I_n))$ converges to zero, and we know that a probability is by definition between 0 and 1, thus meaning the negative log of a probability is strictly positive. By these two facts, that MUST mean that $-\log(P(X_{n+1} | X_n, I_n))$ converges to zero too, thus meaning the model is also learning the external data distribution (remember that the conditional I_n is a way of saying that we are using some specific expert model determined by I_n at that moment).

BY OPTIMIZING FOR $P(X_{n+1}, I_{n+1} | X_n, I_n)$, WE ARE AUTOMATICALLY LEARNING $P(X_{n+1} | X_n, I_n)$, THUS MEANING WE ARE LEARNING THE EXTERNAL DATA DISTRIBUTION AUTOMATICALLY.

This establishes the basis that our model is going to properly learn how to generate text or images or audio, just like a regular non-interiorized model can.

LEARNING THE PROPER INTERIORITY DESPITE IT NOT BEING EXPLICIT

This is a much more difficult question to address. In our model, we are only given explicitly a sequence of external data $X_1, X_2, \dots, X_n$. As such, we have no idea what the proper sequence of corresponding interiorities (virtual experts in this case) are. Suppose thus that we can set an arbitrary j number of possible interiority sequences $I_{(j)1}, I_{(j)2}, \dots, I_{(j)n}$, thus meaning our complete sequence of vectors for a j th possible interiority sequence is $[X_1, I_{(j)1}], [X_2, I_{(j)2}], \dots, [X_n, I_{(j)n}]$.

We can consider for now (we will do something cool with this but just hear me out first) a reinforcement learning model approach to determining what the most optimal sequence of interiorities is.

We can define our j th possible policy as $P_j(I_{n+1} | X_n, I_n)$ (determines the probability distribution of next interiorities). The probability that a given sequence of j th policy decisions occurs is

$\prod_{i=1}^n P_j(I_{i+1} | X_i, I_i) \equiv \Pi_j$. We shall define the total reward for a sequence to be the total probability product of states $\prod_{i=1}^n P_j(X_{i+1}, I_{i+1} | X_i, I_i) \equiv R_j$ (this means our reward is higher for better learning a probability distribution).

In typical reinforcement learning, the goal is optimize for $J = \sum_{k=1}^j \Pi_j R_j$ (essentially a weighted sum of the possible rewards over the possible policy generated sequences). However, due to something called the REINFORCE proof, it turns out that we can actually optimize for J across the large number of j possible policies by just learning through one sequence at a time (meaning we optimize for just $J_1 = \Pi_1 R_1$ first, use this to update the policy $P_1(I_{i+1}|X_i, I_i) \rightarrow P_2(I_{i+1}|X_i, I_i)$, and then sample from said policy to form a new sequence and thus a new $J_2 = \Pi_2 R_2$).

In our design, we are treating our reinforcement learning problem thus as essentially learning the best interiority sequence policy $P(I_{n+1}|X_n, I_n)$ to best realize the reward of best learning $P(X_{n+1}, I_{n+1}|X_n, I_n)$.

Since we defined $\prod_{i=1}^n P_j(I_{i+1}|X_i, I_i) \equiv \Pi_j$ and $\prod_{i=1}^n P_j(X_{i+1}, I_{i+1}|X_i, I_i) \equiv R_j$, this means $J_j = \prod_{i=1}^n P_j(I_{i+1}|X_i, I_i) \prod_{i=1}^n P_j(X_{i+1}, I_{i+1}|X_i, I_i) = \prod_{i=1}^n P_j(I_{i+1}|X_i, I_i) P_j(X_{i+1}, I_{i+1}|X_i, I_i)$.

Maximizing this is the same as minimizing the negative log $-\log(J_j) = \sum_{i=1}^n (-\log(P_j(I_{i+1}|X_i, I_i)) - \log(P_j(X_{i+1}, I_{i+1}|X_i, I_i)))$. By MLE proofs, we know both terms in principle converge to zero.

However, to spice things up, suppose we define $K_j = J_j \cdot \prod_{i=1}^n P(X_{i+1}|I_{i+1}, X_i, I_i)$ and choose to

minimize its negative log $-\log(K_j) = \sum_{i=1}^n (-\log(P_j(X_{i+1}|I_{i+1}, X_i, I_i)) - \log(J_j))$. Once again, by MLE proofs, we know both terms must in principle converge to zero, which means that by optimizing for K_j , we are also optimizing for J_j , thus meaning we are doing our policy reinforcement learning stuff too.

Here is where things get interesting though. $P_j(X_{i+1}, I_{i+1}|X_i, I_i) = P_j(I_{i+1}|X_i, I_i) P_j(X_{i+1}|I_{i+1}, X_i, I_i)$, thus meaning that we can say the following:

$$\begin{aligned} -\log(K_j) &= \sum_{i=1}^n (-\log(P_j(X_{i+1}|I_{i+1}, X_i, I_i)) - \log(P_j(I_{i+1}|X_i, I_i)) - \log(P_j(X_{i+1}, I_{i+1}|X_i, I_i))) \\ -\log(K_j) &= \sum_{i=1}^n (-2\log(P_j(X_{i+1}|I_{i+1}, X_i, I_i)) - 2\log(P_j(I_{i+1}|X_i, I_i))) = -2 \sum_{i=1}^n \log(P_j(X_{i+1}, I_{i+1}|X_i, I_i)) \end{aligned}$$

This is just a scaled factor way of showing that minimizing $-\log(K_j)$ is identical to minimizing $-\sum_{i=1}^n \log(P_j(X_{i+1}, I_{i+1}|X_i, I_i))$. But, and this is the big thing, we know that minimizing $-\log(K_j)$ must minimize $-\log(J_j)$ by MLE proofs AND that minimizing $-\sum_{i=1}^n \log(P_j(X_{i+1}, I_{i+1}|X_i, I_i))$ is just the same basic negative log likelihood method used to learn $P_j(X_{i+1}, I_{i+1}|X_i, I_i)$.

THUS, LEARNING TO OPTIMIZE FOR $P_j(X_{i+1}, I_{i+1}|X_i, I_i)$ ORGANICALLY CAUSES ONE TO LEARN THE OPTIMAL POLICY $P_j(I_{i+1}|X_i, I_i)$ TOO IN A WAY THAT IS SCALABLE.

What we thus see is that by simply incorporating an ontology of interiority (learning $P(X_{i+1}, I_{i+1}|X_i, I_i)$ instead of $P(X_{i+1}|X_i)$), we are capable of creating a learning methodology from which external data learning AND interior policy reinforcement learning **HAPPEN ORGANICALLY AND SIMULTANEOUSLY**.

BASIC LEARNING LOOP

initialize external data vectors X_1 to X_n in sequence

generate initial interiority $I_{1(1)}$ via $[X_{arbitrary}, I_{1(1)}] = f_x([\times, \times])$ ($\times$ just means initialization token)

append to make $V_{1(1)} = [X_1, I_{1(1)}]$

for i in range 2 to n:

$$I_{i(1)} = f_{I_{i-1(1)}}(V_{i-1(1)})$$

$$V_{i(1)} = [X_i, I_{i(1)}]$$

repeat until we reach satisfactory convergence (represent via iteration j):

perform model learning to predict V_i from $V_{j < i}$ for k epochs (set k to whatever you want)

NOTE THIS MEANS WE WANT TO LEARN SUCH THAT $f_{I_{i(j)}}(V_{i(j)}) \rightarrow V_{i+1(j)}$

$f_{I_{i(j)}}$ is the selected virtual expert

calculate and optimize for negative log likelihood to learn $P_j(X_{i+1}, I_{i+1(j)}|X_i, I_{i(j)})$

generate new interiority $I_{1(j+1)}$ via $[X_{arbitrary}, I_{1(j+1)}] = f_x([\times, \times])$

append to make $V_{1(j+1)} = [X_1, I_{1(j+1)}]$

for i in range 2 to n:

$$I_{i(j+1)} = f_{I_{i-1(j+1)}}(V_{i-1(j+1)})$$

$$V_{i(j+1)} = [X_i, I_{i(j+1)}]$$